

PHASE – 1

Software Requirements

Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux (supports cross-platform operations).
- **Node.js (v16 or above):** Runtime ecosystem running server logic and routing infrastructure.
- **npm (v8 or above):** Node package manager to control operational dependencies.
- **Gemini AI SDK:** Integration using @google/generative-ai for natural language insights.
- **MongoDB:** Document NoSQL engine storing users, posts, categories, comments, and analytics metrics.
- **Postman:** API verification toolkit to assert schema validation responses across administrative route shields.
- **Code Editor:** Visual Studio Code or similar IDE environment.

Technologies Used

- Visual Studio Code – write and Runs the Backend code.
- Node.js – Runs the backend server and application logic.
- Express.js – Handles the REST APIs, routes, requests, and responses.
- MongoDB – Stores users and favorite location data.
- Mongoose – Connects the Node.js application with MongoDB and manages schemas.
- OpenWeatherMap – Provides real-time weather data.
- Google Gemini AI – Generates weather summaries and personalized recommendations.
- JWT – Provides user authentication and protects secured routes.
- bcrypt – Hashes user passwords securely.
- Postman – Used to test the REST APIs.

Short Explanation

“AI WeatherWise is a Node.js and Express.js based RESTful backend application that uses MongoDB for data storage, OpenWeatherMap for real-time weather data, and Google Gemini AI to generate intelligent weather summaries and personalized recommendations.”